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Section : 08

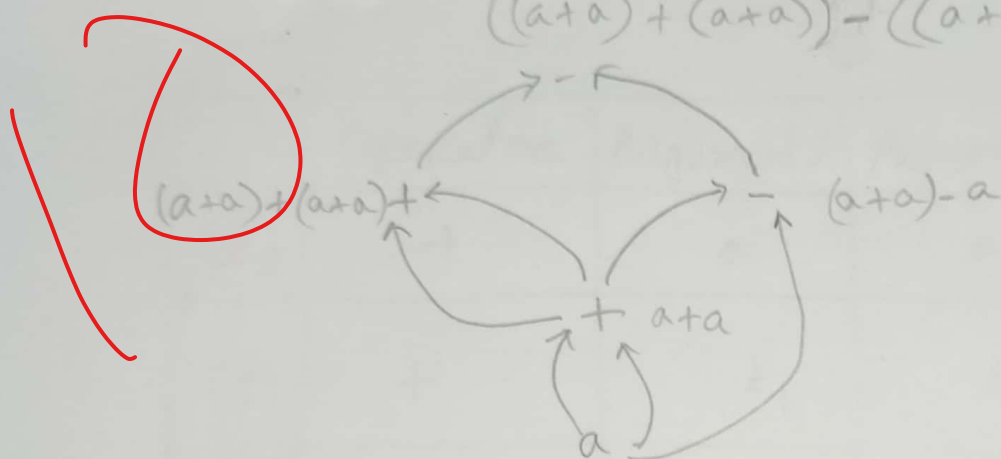
Course code : CSE 420

Final

(1)

Ans: to the Question No: 3

$$((a+a) + (a+a)) - ((a+a) - a)$$



(a)

The DAG represents $((a+a) + (a+a)) - ((a+a) - a)$

The 3 address code:

$$\begin{aligned}
 (0) \quad & t_1 = a + a \\
 (1) \quad & t_2 = t_1 + t_1 \\
 (2) \quad & t_3 = t_1 - a \\
 (3) \quad & t_4 = t_2 - t_3
 \end{aligned}$$

$$(0) \quad t_1 = a + a$$

$$(1) \quad t_2 = t_1 + t_1$$

$$(2) \quad t_3 = t_1 - a$$

$$(3) \quad t_4 = t_2 - t_3$$

⑥ Quadruples:

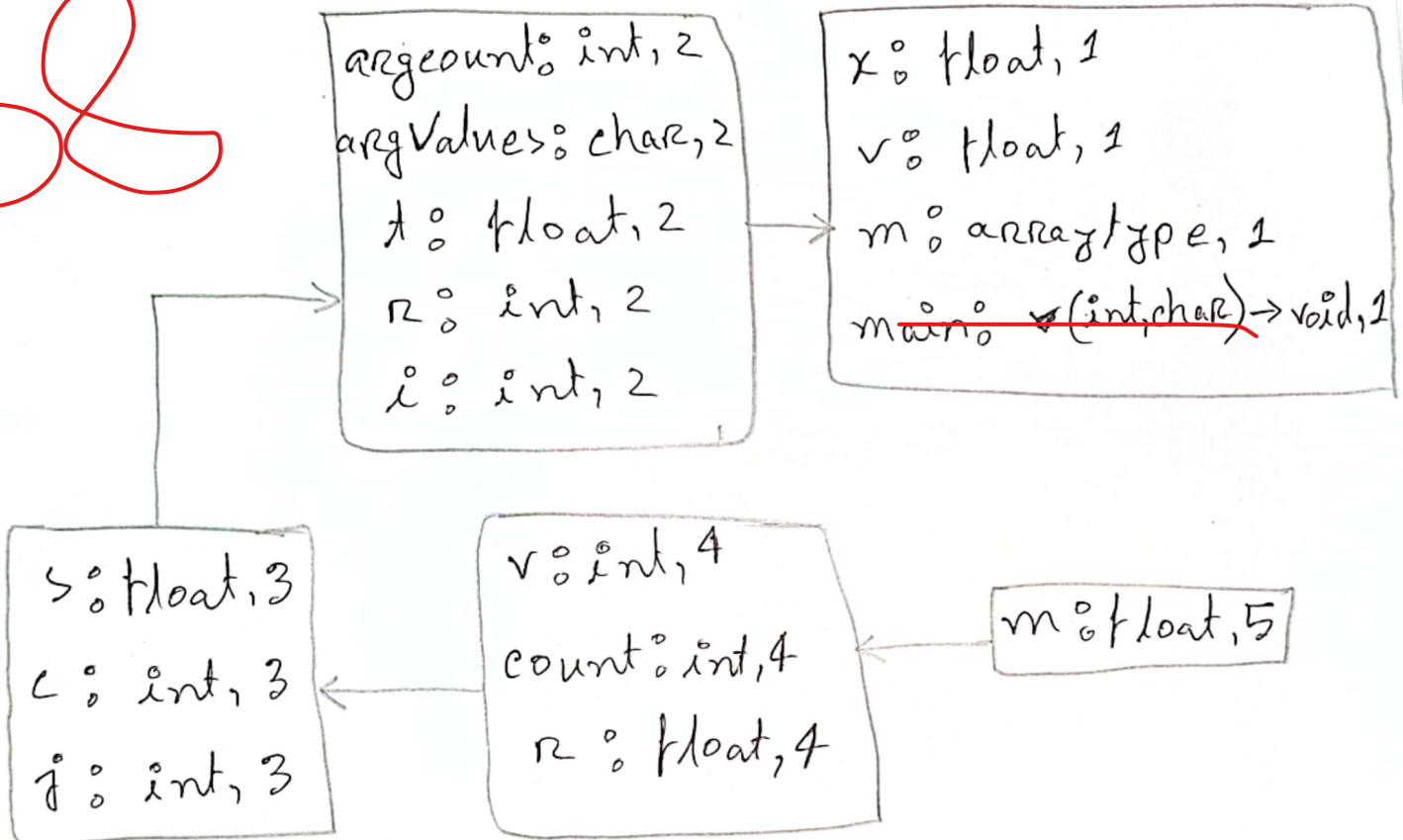
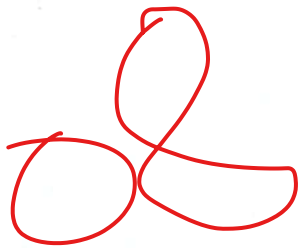
	Operator	Argument 1	Argument 2	Result
(0)	+	a	a	t_1
(1)	+	t_1	t_1	t_2
(2)	-	t_1	a	t_3
(3)	+ -	t_2	t_3	t_4

Triples:

	Operator	Argument 1	Argument 2
(0)	+	a	a
(1)	+	(0)	(0)
(2)	-	(0)	a
(3)	+ -	(1)	(2)

Ans: to the Question No: 2

By using list of hash tables,

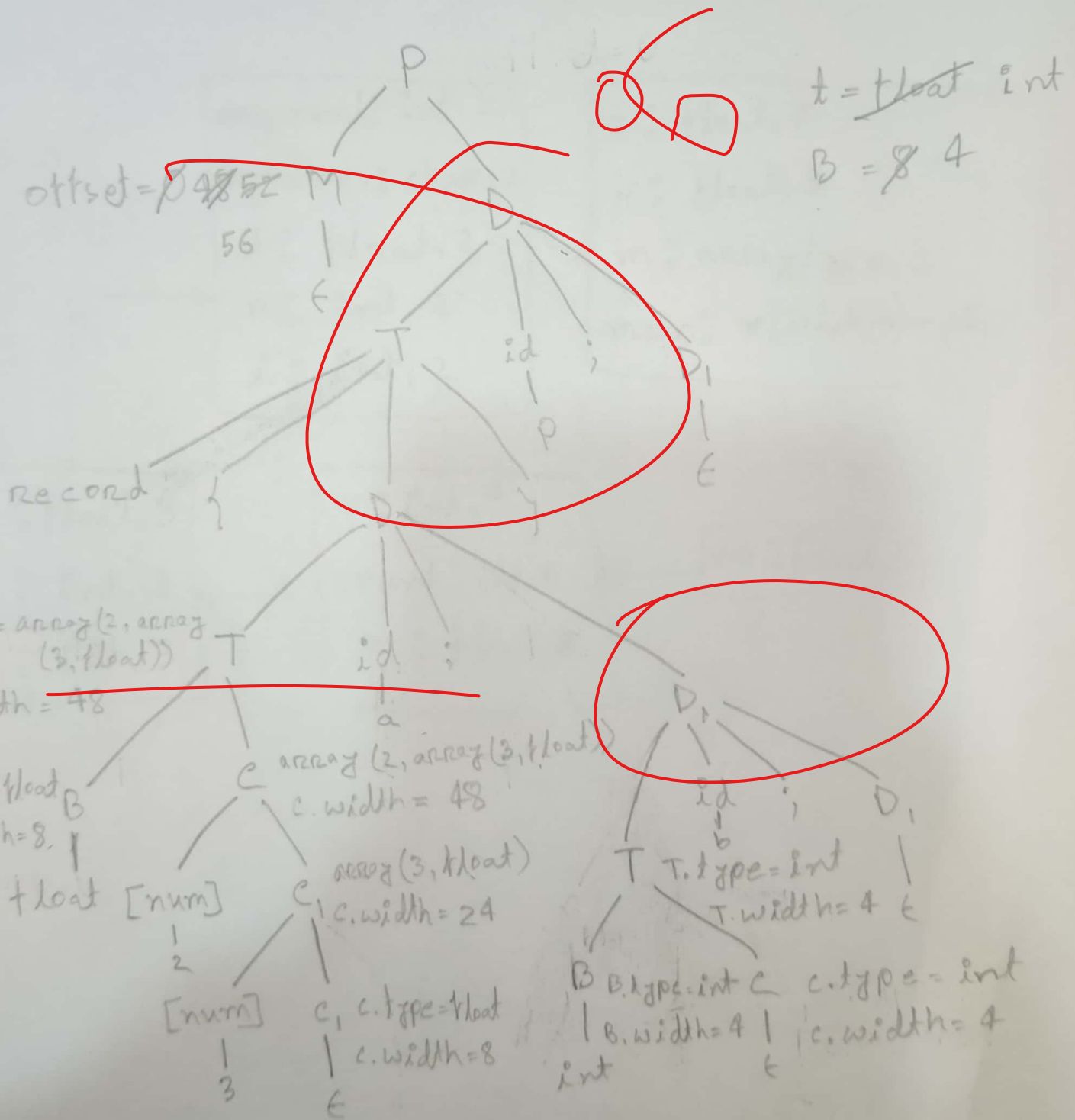


offset = 04852 M

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Diagram showing a tree structure with a root node 'P' and children 'M' and 'D'. A red line is drawn across the diagram.

~~$t = t_{\text{float}}$~~ int
B = 8 4



Symbol table:

Variable	Type	offset
a	array(2, array(3, float))	0
b	int	48
P	int	52